The Magic Number Conjecture for the Rank-Two $m = 4$ Amplituhedron

A simplicial reduction of global tile relations to finite local data

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Abstract

For positive external data $Z \in \mathbb{R}^{n\times 6}$, the rank-two $m = 4$ amplituhedron $\mathcal{A}_{n,2,4}(Z)$ is the image of $\mathrm{Gr}_{\geq 0}(2, n)$ under $C \mapsto CZ$. We prove that every complete multiplicity-one all-Z positroid tiling of $\mathcal{A}_{n,2,4}$ has

$$M(2, n-6, 2) = \frac{1}{n-3}\binom{n-3}{2}\binom{n-3}{3} = \frac{(n-3)(n-4)^2(n-5)}{12}$$

tiles, for every admissible $n \geq 6$.

The proof separates local and global issues. Let V_n be the real vector space generated by all eight-dimensional positroid source cells whose amplituhedron maps are injective for every positive Z. Let L_n be the span of the geometric codimension-one parent relations, and let K_n be the kernel of the family of all-Z tile-indicator maps, modulo semialgebraic sets of dimension at most seven. A finite rank-two classification proves that every local parent relation is balanced. The main global theorem is

$$K_n = L_n \qquad (n \geq 6).$$

Its proof uses signed adjacent-column amalgamation as face maps and zero insertion as degeneracy maps. These operations make V, L, K, and K/L into simplicial real vector spaces. Normalization then reduces the infinite problem to finitely many support cores. The normalized quotient has dimensions $1, 9, 19, 12, 0, 0$ in source label numbers $6, \ldots, 11$ and vanishes thereafter by a support bound. Since $K_9 = L_9$ is certified exactly, exactness and faithfulness of simplicial normalization force $K/L = 0$ in every degree.

The proof is computer-assisted only in finite classification and exact-rank certificates; all matrix computations are over $\mathbb{Q}$ or $\mathbb{Z}$ and are accompanied by exact rank certificates, including nonzero minors and upper-rank certificates where needed. No floating-point rank calculation, exhaustive tiling enumeration, or fixed-Z extrapolation is used.

Status of this draft. This manuscript presents a computer-assisted proof with an accompanying exact certificate supplement. It has not yet undergone external peer review. Each computational claim below carries a stable certificate ID; Appendix A describes the literal data and verification conditions. The supplement's `CERTIFICATE_INDEX.md` maps these IDs to files and replay commands. The supplement should accompany every circulated version.

1 Introduction

The amplituhedron $\mathcal{A}_{n,k,m}$ was introduced as a positive-geometric model for scattering amplitudes and has since developed a substantial combinatorial theory. Karp–Williams–Zhang conjectured the

existence, for even m, of a positroid tiling of $\mathcal{A}_{n,k,m}$ of cardinality

$$M\left(k, n-k-m, \frac{m}{2}\right),$$

the MacMahon number counting plane partitions in a $k \times (n-k-m) \times (m/2)$ box [4, Conjecture 8.1]. The assertion that every tiling has this cardinality is formulated in [3, Conjecture 3.9]. Parisi, Sherman-Bennett, Tessler, and Williams proved the conjecture for $m = 2$ using T-duality and Parke–Taylor weights [7]. For $m = 4$, Even-Zohar, Lakrec, Parisi, Tessler, Sherman-Bennett, and Williams proved that all iterated BCFW constructions give genuine tilings [5]; a companion paper exhibited the non-BCFW spurion tile [6]. Thus the cardinality problem for arbitrary positroid tilings is strictly stronger than a statement about BCFW tilings alone.

The purpose of this paper is to settle the first higher-m rank slice: $k = 2$, $m = 4$, and arbitrary n. The main result is the following.

Theorem 1.1 (Magic Number for $\mathcal{A}_{n,2,4}$)**.** *For every $n \geq 6$, every complete multiplicity-one all-Z positroid tiling $\mathcal{T}$ of $\mathcal{A}_{n,2,4}$ has*

$$|\mathcal{T}| = M(2, n-6, 2) = \frac{1}{n-3}\binom{n-3}{2}\binom{n-3}{3} = \frac{(n-3)(n-4)^2(n-5)}{12}.$$

The proof is organized around relations among tile indicators. For each n we define a free tile space V_n, the span L_n of local relations coming from nine-dimensional positroid parents, and the full all-Z indicator kernel K_n. The local theorem proves that every vector in L_n has coefficient sum zero. The global problem is therefore reduced to proving

$$K_n = L_n.$$

At rank two, this equality admits an unexpectedly finite proof. Adjacent column amalgamation and zero insertion turn the system into a simplicial vector space. Passing to the normalized quotient discards all generators obtained by inserting loops. A classification of all rank-two $m = 4$ tiles shows that a tile has at most ten nonloop labels. Consequently no new normalized generators can occur after a finite degree. Two exact finite-rank computations at label numbers 10 and 11, together with the previously certified equality $K_9 = L_9$, close the global argument.

Outline

Section 2 fixes conventions. Section 3 gives the finite membership-core theorem. Section 4 states the complete local parent theorem and explains why all local relations are balanced. Section 5 proves geometric descent under adjacent amalgamation. Section 6 establishes the simplicial identities. Section 7 gives the finite normalized-rank calculation, and Section 8 proves $K_n = L_n$ for all n. The Magic Number formula is deduced in Section 9. Appendices summarize the exact finite certificates and reproducibility requirements.

2 Positive Grassmannians, tiles, and indicator relations

2.1 Rank-two positroid cells

A point of $\mathrm{Gr}(2, n)$ is represented by a full-rank $2 \times n$ matrix $C = (c_1, \ldots, c_n)$ modulo left multiplication by GL_2. Write $\Delta_{ij}(C)$ for the ordered 2×2 Plücker coordinate. The totally nonnegative Grassmannian is

$$\mathrm{Gr}_{\geq 0}(2, n) = \{C : \Delta_{ij}(C) \geq 0 \text{ for } i < j\}.$$

Its positroid stratification is recalled from Postnikov [1]. In rank two a cell is determined by its loops together with cyclically ordered parallel classes among the nonloop columns. If q is the number of nonloop labels and s the number of cyclic parallel classes, then

$$\dim S = q + s - 4. \tag{1}$$

2.2 The amplituhedron map

Fix a positive matrix $Z \in \mathbb{R}^{n\times 6}$, meaning that all increasing 6×6 minors are positive. The rank-two $m = 4$ amplituhedron map is

$$f_Z : \mathrm{Gr}_{\geq 0}(2, n) \longrightarrow \mathrm{Gr}(2, 6), \qquad C \longmapsto CZ,$$

with CZ interpreted by row span. Its image is $\mathcal{A}_{n,2,4}(Z)$, which has dimension 8.

Definition 2.1 (All-Z tile)**.** *An eight-dimensional positroid cell $S \subset \mathrm{Gr}_{\geq 0}(2, n)$ is an* all-Z tile source *if $f_Z|_S$ is injective for every positive $n \times 6$ matrix Z. We write*

$$T_S(Z) = f_Z(\overline{S}) \subset \mathcal{A}_{n,2,4}(Z)$$

for its closed tile image.

The all-Z convention is essential throughout. A cell that is injective for one chosen positive matrix but not uniformly in Z is not included in the tile universe used here.

Definition 2.2 (Complete multiplicity-one all-Z tiling)**.** *A finite collection $\mathcal{T}$ of distinct all-Z tile sources is a* complete multiplicity-one all-Z positroid tiling *if the collection is independent of Z and, for every positive $n \times 6$ matrix Z,*

$$\mathcal{A}_{n,2,4}(Z) = \bigcup_{S\in\mathcal{T}} T_S(Z), \qquad f_Z(S) \cap f_Z(S') = \varnothing \quad (S \neq S').$$

Thus the closed tiles cover the entire amplituhedron, and their open images are pairwise disjoint. In particular, $\sum_{S\in\mathcal{T}} \mathbf{1}_{T_S(Z)} = \mathbf{1}_{\mathcal{A}_{n,2,4}(Z)}$ modulo semialgebraic sets of dimension at most seven; common tile boundaries are allowed, and no face-to-face condition is imposed.

2.3 Relations modulo lower-dimensional sets

Let $\mathcal{U}_n$ denote the finite set of all all-Z tile sources and put

$$V_n = \mathbb{R}^{\mathcal{U}_n} = \bigoplus_{S\in\mathcal{U}_n} \mathbb{R}e_S.$$

For a fixed positive Z, two indicator functions are identified when they differ only on a semialgebraic subset of dimension at most 7. Define the uniform all-Z indicator kernel

$$K_n = \left\{ c \in V_n : \sum_{S\in\mathcal{U}_n} c_S \mathbf{1}_{T_S(Z)} = 0 \pmod{\dim \leq 7} \text{ for every positive } Z \right\}. \tag{2}$$

The coefficients c_S are independent of Y and Z.

A complete multiplicity-one all-Z tiling $\mathcal{T}$ defines a binary vector

$$\chi_{\mathcal{T}} = \sum_{S\in\mathcal{T}} e_S \in V_n.$$

If $\mathcal{T}$ and $\mathcal{T}'$ tile the same amplituhedron, then

$$\chi_{\mathcal{T}} - \chi_{\mathcal{T}'} \in K_n. \tag{3}$$

Thus controlling K_n controls all tiling differences without requiring flip connectivity or enumeration of all tilings.

3 The complete rank-two membership classification

The global argument will use the fact that all tile sources have bounded nonloop support. We therefore record the complete rank-two classification.

Theorem 3.1 (Membership-core theorem)**.** *Let $S \subset \mathrm{Gr}_{\geq 0}(2,n)$ be an eight-dimensional positroid cell. Let q be its number of nonloop labels and let $b_1, \dots, b_s$ be the cyclic parallel-class sizes. Then S is an all-Z tile source if and only if either*

$$\max_i b_i = q - 5, \tag{4}$$

or

$$(q; s; b_1, b_2, b_3) = (9; 3; 3, 3, 3).$$

Up to signed dihedral symmetry and insertion of loops, there are exactly eleven admissible loopless core types: ten satisfying (4) *and the* $(3,3,3)$ *spurion.*

Proof. By (1), dimension eight gives $q + s = 12$, hence

$$(q,s) = (6,6), (7,5), (8,4), (9,3), (10,2).$$

If a parallel class D has at most four complementary nonloop labels, then a nonzero source row functional annihilating the direction of D produces a nonzero image vector supported on at most four fixed external rows. Therefore the image plane meets a fixed external t-space with $t \leq 4$. The corresponding incidence locus in $\mathrm{Gr}(2,6)$ has dimension at most $t+3 \leq 7$, so the eight-dimensional cell cannot map injectively to an open eight-dimensional image.

If no class has exactly five complementary labels and no class has at most four complementary labels, the five possible pairs (q,s) force only two remaining profiles: $(2,2,2,2)$ and $(3,3,3)$. The former is the four-mass cell and is not all-Z injective; see Galashin–Lam [3] and the exact two-preimage certificate in the supplementary computation. The latter is the spurion, whose all-Z injectivity is the published input stated in [6, Section 5.1]. The explicit four-mass data are `MEM-FOUR-MASS`; the spurion inverse fixtures are `MEM-SPURION`. The latter fixtures check the inverse implementation; they do not replace the all-Z injectivity input.

It remains to prove injectivity for the complement-five family. Delete loops and cyclically cut so that a distinguished class D is last and its five complementary labels occur first. If $A_1, \dots, A_\ell$ are the successive complementary classes and $r = |D|$, then $\ell + r = 6$ and every open source point admits the chart

$$C = \begin{pmatrix} a_1 & \cdots & a_5 & 0 & \cdots & 0 \\ t_{j(1)}a_1 & \cdots & t_{j(5)}a_5 & d_1 & \cdots & d_r \end{pmatrix}, \qquad a_i, d_h > 0, \quad t_1 < \cdots < t_\ell.$$

Let $H = \mathrm{span}(Z_1, \dots, Z_5)$ and $h(X) = \det(Z_1, \dots, Z_5, X)$. The first image row lies in H, while the second has positive h-value, so $Y \cap H$ is a unique line. Its positive ray uniquely determines $a_1, \dots, a_5$ up to common scale. Put

$$V_j = \sum_{i \in A_j} a_i Z_i.$$

The six rows

$$(V_1, \dots, V_\ell, Z_{D_1}, \dots, Z_{D_r})$$

are independent because their determinant is a positive sum of increasing 6×6 minors of Z. Expanding any second image row of positive h-value in this basis recovers all slopes t_j and all d_h

uniquely. Thus the source row space is uniquely recovered. This proves all-Z injectivity for all ten complement-five cores.

The finite composition list is then obtained from the five possibilities $q + s = 12$ up to cyclic rotation and reversal. The eleven admissible core types are displayed in Table 1. Loop insertion and signed dihedral transport preserve the entire source/image problem, so the classification holds for arbitrary n. The complete labeled catalogue, its dihedral orbits, and the finite composition enumeration are supplied as `MEM-CORES`. □

q	number of loopless cells	admissible class-size cores
6	1	$(1,1,1,1,1,1)$
7	14	$(1,1,1,2,2)$, $(1,1,2,1,2)$
8	36	$(1,1,3,3)$, $(1,2,2,3)$, $(1,2,3,2)$, $(1,3,1,3)$
9	30	$(1,4,4)$, $(2,3,4)$, $(3,3,3)$
10	5	$(5,5)$

Table 1: The eleven admissible loopless membership cores. The $(3,3,3)$ profile is the spurion; the other ten are complement-five. The labeled counts and orbit data are certificate `MEM-CORES`.

Corollary 3.2 (Support bound and tile-source count)**.** *Every all-Z tile source has at most ten nonloop labels. Moreover*

$$|\mathcal{U}_n| = \binom{n}{6} + 14\binom{n}{7} + 36\binom{n}{8} + 30\binom{n}{9} + 5\binom{n}{10}. \tag{5}$$

The counts in (5) are replayed by `MEM-CORES`; `SUPPORT-CUTOFF` checks their finite support range and the resulting normalized cutoff used below.

4 The local parent relation theorem

Let P be a nine-dimensional positroid cell and let

$$\mathcal{F}(P) = \{S : \dim S = 8,\ S \lessdot P,\ S \in \mathcal{U}_n\}$$

be its admissible codimension-one source facets. Define

$$\mathcal{R}_{\mathrm{all}}(P) = \left\{ c \in \mathbb{R}^{\mathcal{F}(P)} : \sum_{S \in \mathcal{F}(P)} c_S \mathbf{1}_{T_S(Z)} = 0 \pmod{\dim \le 7} \text{ for every positive } Z \right\}.$$

Let $L_n \subset V_n$ be the span of all such geometric one-parent relations, together with their loop and signed-dihedral transports.

Theorem 4.1 (Complete one-parent theorem)**.** *For every nine-dimensional rank-two positroid parent P and every $n \ge 7$, the space $\mathcal{R}_{\mathrm{all}}(P)$ is either zero or the one-dimensional span of a primitive balanced vector. Up to loop deletion and dihedral relabeling, the* 29 *loopless parent cores split as follows:*

sector	*number of cores*	*admissible facets*	*relation space*
collapsed parent image	*12*	*0*	0
complement-five circuit	*10*	*6*	$\mathbb{R}\varepsilon$, $3 \leftrightarrow 3$
reflection/spurion	*5*	*10*	$\mathbb{R}\varepsilon$, $5 \leftrightarrow 5$
hostile four-mass	*2*	*8*	0

In particular, every $r \in L_n$ satisfies

$$\sum_{S \in \mathcal{U}_n} r_S = 0. \tag{6}$$

Proof structure. For rank two, a nine-dimensional parent has $q + s = 13$. Up to cyclic rotation and reversal this yields 29 loopless core types. The source facets of every core are enumerated exactly and are classified using Theorem 3.1: each facet is either an all-Z tile, collapsed by the class-complement obstruction, or the noninjective four-mass type.

For ten cores a distinguished class has exactly five complementary labels. The positive fiber is an affine interval whose six actual wall functions have alternating derivatives, obtained from the Gale vector of a positive 7×6 matrix. This gives a multiplicity-one $3 \leftrightarrow 3$ relation. Four further cores admit an orientation-reversing source reflection paired with an orientation-preserving target action; the ten facets therefore pair with opposite image signs, giving $5 \leftrightarrow 5$. The remaining balanced core is the repaired spurion pattern, whose wall-sign sum is zero by an ordered-minor/Plücker calculation.

It remains to show that no additional unbalanced relation exists on the same facet set. For the six-wall cores, opposite-side open overlaps form $K_{3,3}$, forcing every coefficient relation to be a scalar multiple of the known balanced vector. For each ten-wall core, nine exact rational open-overlap witnesses form a spanning tree on the ten facets; the nine edge equations have rank nine and therefore one-dimensional kernel. For the two hostile cores, eight exact target neighborhoods produce an 8×8 tile-indicator matrix of determinant 1, so their relation spaces are zero. All determinants and membership/nonmembership decisions are checked by exact replay. The complete catalogue is `LOCAL-PARENTS`; its literal primitive vectors and their zero coefficient sums are `LOCAL-RELATIONS`. The 45 overlap witnesses and five rank-nine matrices are `LOCAL-TEN-WALL`, and the two hostile matrices are `LOCAL-HOSTILE-8` and `LOCAL-HOSTILE-9`. Loop insertion and signed dihedral transport extend the finite core statement to all n. □

Remark 4.2 (Computer-assisted scope)**.** *The finite core enumeration, the ten-wall spanning-tree certificates, and the two hostile determinant certificates are exact finite computations. No floating-point geometry is used. The symbolic complement-five family and the transport arguments are uniform in positive Z.*

5 Adjacent amalgamation and geometric descent

We now introduce the global structure that makes the arbitrary-n problem finite. For a source matrix $C = (c_1, \dots, c_n)$ define ordinary adjacent amalgamation by replacing two adjacent columns by their sum. The cyclic wrap requires the rank-two sign twist.

For $0 \le i < n - 1$, set

$$d_i(c_1, \dots, c_n) = (c_1, \dots, c_{i+1} + c_{i+2}, \dots, c_n), \tag{7}$$

and set

$$d_{n-1}(c_1, \dots, c_n) = (c_1 - c_n, c_2, \dots, c_{n-1}). \tag{8}$$

On the free tile space, retain the resulting source label only when it is again an admissible eight-dimensional tile source; otherwise define the image to be zero.

Theorem 5.1 (Geometric amalgamation)**.** *For every $n \geq 7$, the maps $d_i : V_n \to V_{n-1}$ have the following properties.*

(i) Each source face has a single exact positroid-cell image. If source dimension is preserved and the output is admissible, the map between the open source cells is a positive rational isomorphism.

(ii) If an eight-dimensional output is discarded, then it is collapsed, not four-mass; its target image has dimension at most seven for every positive downstairs external matrix.

(iii) The maps descend to all-Z indicator kernels:

$$d_i(K_n) \subseteq K_{n-1}.$$

(iv) They descend to the local relation spans:

$$d_i(L_n) \subseteq L_{n-1}.$$

Proof. We give the ordinary last-pair case; all other ordinary positions follow by signed cyclic transport, and the wrap is the same geometric operation in twisted cyclic coordinates.

Write the last two source columns as a, b and their amalgamation as $w = a + b$. Every output ordered minor is either an unchanged source minor or

$$\Delta_{i,w} = \Delta_{i,n-1} + \Delta_{i,n} \qquad (i < n-1).$$

Since all summands are nonnegative, there is no cancellation; the full source zero pattern therefore determines a unique output positroid label. Rank loss on the nonnegative Grassmannian can occur only at the coordinate two-plane supported on the merged pair. Indeed, if every minor involving w vanishes, the nonnegative summands vanish separately, and any surviving outside column would force all columns to be parallel, contradicting rank two.

A direct case analysis on loops, same-class amalgamation, and distinct parallel classes shows that every dimension-preserving nonloop face admits a rational inverse. Geometrically, the output direction w lies strictly between the two surviving old directions; its positive decomposition in those two rays uniquely recovers the original pair up to source row-space gauge. Faces involving a disappearing singleton or same-class pair lower dimension and are treated only as lower-dimensional outputs.

By Theorem 3.1, a dimension-preserving nonloop face creates a singleton class, so an inadmissible eight-dimensional output cannot be the four-mass profile $(2, 2, 2, 2)$. Every other excluded eight-cell has a class with at most four complementary nonloops and is therefore collapsed by the incidence-dimension argument used in Theorem 3.1.

Fix an arbitrary positive downstairs external matrix W. It can be realized as a positive degeneration of upstairs external matrices $Z(\epsilon)$ in which the relevant adjacent rows coalesce as $\epsilon \to 0^+$. One may choose the perturbation so that every increasing six-minor remains positive for sufficiently small ϵ.

The only source rank-loss limit is the coordinate plane supported on the merged pair. Every limiting target from this locus contains the duplicated external row and therefore lies in a Schubert locus of dimension four. All collapsed or lower-dimensional output cells contribute target sets

of dimension at most seven. Away from the finite union of these exceptional sets and source-boundary/critical images, surviving tile indicators are locally constant under the degeneration: membership follows from the rational source isomorphism and the parameter-dependent inverse function theorem, while closed nonmembership follows by compactness of source closures.

Now let $r \in K_n$. On a generic downstairs open neighborhood, every upstairs tile indicator appearing in r stabilizes to the corresponding downstairs indicator, with multiplicity retained. If $d_i(r)$ were nonzero on that neighborhood, the same nonzero value would persist for all small positive ϵ, contradicting the all-Z relation defining r. Hence $d_i(K_n) \subseteq K_{n-1}$.

Finally take a primitive one-parent relation. The exact image of its nine-dimensional parent is either nine-dimensional, eight-dimensional, or smaller. In the first case the surviving facet relation belongs to the already classified one-parent kernel of the downstairs parent. In the second case all surviving facets have the same eight-dimensional output label, and the resulting scalar relation must be zero on its nonempty open image. In the last case every surviving output is lower-dimensional. Thus $d_i(L_n) \subseteq L_{n-1}$ without using the global equality to be proved. □

6 The twisted simplicial structure

For $0 \le i < n$, let $s_i : V_n \to V_{n+1}$ insert a zero source column after c_{i+1}. This is loop insertion. On source closures,

$$(j_I C)Z = CZ_I, \tag{9}$$

so zero insertion preserves all-Z tile relations and local parent relations exactly.

Theorem 6.1 (Simplicial structure)**.** *Set $V_n = L_n = K_n = 0$ for $n \le 5$. With simplicial degree $p = n - 1$, the spaces*

$$V_{p+1}, \qquad L_{p+1}, \qquad K_{p+1}, \qquad \mathcal{Q}_p := V_{p+1}/L_{p+1}, \qquad \mathcal{H}_p := K_{p+1}/L_{p+1}$$

form simplicial real vector spaces, and $\mathcal{H}$ is a simplicial subobject of $\mathcal{Q}$.

Proof. At the raw column level, the face and degeneracy maps satisfy

$$d_i d_j = d_{j-1} d_i \qquad (i < j), \tag{10}$$

$$d_i s_j = \begin{cases} s_{j-1} d_i, & i < j, \\ \mathrm{id}, & i = j \text{ or } i = j+1, \\ s_j d_{i-1}, & i > j+1, \end{cases} \tag{11}$$

$$s_i s_j = s_{j+1} s_i \qquad (i \le j). \tag{12}$$

For non-wrap faces these are associativity and insertion identities. The two endpoint computations involving the wrap are

$$(x_0 - x_{n-1}) + x_1 = (x_0 + x_1) - x_{n-1},$$

and

$$(x_0 - x_{n-1}) - x_{n-2} = x_0 - (x_{n-2} + x_{n-1}).$$

The face–degeneracy identities at the cyclic boundary are checked in the same signed convention.

It remains to justify truncation: a raw intermediate that has been sent to zero must never become an admissible eight-cell later on one side of an identity. Rank loss and source-dimension drop cannot be repaired by later faces or zero insertion. If the discarded intermediate is an inadmissible eight-cell, Theorem 3.1 and the singleton created by a dimension-preserving nonloop amalgamation

imply that it is collapsed, not four-mass. Its class-complement obstruction persists under every subsequent dimension-preserving face: amalgamating away from the obstructing class decreases the complement, while amalgamating the obstructing class decreases both its size and total support. Hence a discarded intermediate remains discarded. The raw equalities therefore descend to the truncated free tile spaces.

By Theorem 5.1, faces preserve K and L. By (9), degeneracies preserve K and L. The simplicial identities consequently restrict to these subspaces and descend to their quotients. □

7 Normalization and the finite quotient

For a simplicial real vector space X, let $N(X)$ denote its normalized chain complex. We use the standard facts that $N(X)_p$ is canonically isomorphic to the quotient of X_p by the span of degeneracies from degree $p-1$, that normalization is exact, and that it detects the zero simplicial object. These are standard consequences of Dold–Kan; see the Stacks Project [8, 9, 10].

Since s_i inserts a loop in one of positions $2, \ldots, n$, normalization of the quotient $Q_n := V_n/L_n$ gives

$$N(\mathcal{Q})_{n-1} \cong V_n \Big/ \left(L_n + \operatorname{span}\{e_S : S \text{ has a loop in } \{2, \ldots, n\}\}\right). \tag{13}$$

A tile whose only loop is at position 1 survives this quotient.

The support bound in Corollary 3.2 now makes the normalized problem finite. The surviving columns at label number n are exactly the loopless n-label tiles together with the $(n-1)$-label loopless tiles inserted with their unique loop at position 1. The only projected parent rows that can be nonzero come from loopless parents or parents with their unique loop at position 1.

Proposition 7.1 (Exact normalized ranks)**.** *The normalized quotients have the following exact finite presentations:*

n	*surviving tile columns*	*projected parent rows*	*row rank*	$\dim N(\mathcal{Q})_{n-1}$
6	*1*	*0*	*0*	*1*
7	*15*	*7*	*6*	*9*
8	*50*	*47*	*31*	*19*
9	*66*	*121*	*54*	*12*
10	*35*	*141*	*35*	*0*
11	*5*	*46*	*5*	*0*

The displayed maximal nonzero minors can be chosen with determinants

$$1, -1, 1, -1, -1, 1$$

in order, with the first entry denoting the determinant 1 *of the* 0×0 *minor of the rank-zero matrix. The six rows of the table are certificates* `NORM-6`*,* `NORM-7`*,* `NORM-8`*,* `NORM-9`*,* `NORM-10`*, and* `NORM-11`*. For* $n \geq 12$ *there are no normalized tile generators (*`SUPPORT-CUTOFF`*). Consequently*

$$N(\mathcal{Q})_p = 0 \qquad (p \geq 9). \tag{14}$$

Proof. The column and row sets are determined exhaustively from the eleven membership cores and the 29 parent cores. Rows from parents carrying a loop in positions $2, \ldots, n$ vanish after projection because every facet retains that loop. The remaining rows are literal transports of the finite parent prototypes.

All ranks are computed over $\mathbb{Q}$. For $n = 10$ the selected 35×35 submatrix has determinant -1, and for $n = 11$ the selected 5×5 submatrix has determinant 1. These give full column rank. For every displayed rank, the supplement also supplies exact rational reduced rows and pivot data; checking that every reconstructed row lies in their span gives the upper bound, while the selected minor gives the lower bound. This includes the non-full-rank cases $n = 7, 8, 9$. The independent construction and its shared arithmetic dependencies are specified in Appendix B. Finally, every tile has at most ten nonloop labels by Corollary 3.2; for $n \geq 12$, every tile therefore has a loop in positions $2, \ldots, n$ and dies in the normalized quotient. □

Remark 7.2. *Only the two high-degree zero statements at $n = 10, 11$ and the support cutoff are needed for the all-n vanishing argument. The lower normalized dimensions are useful for the rank formula below but are not an extrapolation hypothesis.*

8 Uniform global generation

The finite base case at $n = 9$ is the last ingredient. It is obtained by a complete exact indicator-kernel certificate using the 942 all-Z tile sources at $n = 9$ (`N9-UNIVERSE`). The 693×942 local relation matrix has rank 470 (`N9-LOCAL-RANK`). An exact 472×942 open-neighborhood indicator matrix has rank 472, with a 472×472 minor of determinant 4, and annihilates every local relation (`N9-INDICATOR-RANK`, `N9-ANNIHILATION`). A separate implementation reconstructs all 444,624 membership decisions from the 472 exact target records, including the spurion inverse (`N9-INDICATORS`). Therefore `N9-KERNEL` gives

$$K_9 = L_9. \tag{15}$$

We isolate the resulting infinite statement.

Theorem 8.1 (Uniform indicator generation)**.** *For every $n \geq 6$,*

$$K_n = L_n.$$

Proof. Let

$$\mathcal{H}_p = K_{p+1}/L_{p+1}, \qquad \mathcal{Q}_p = V_{p+1}/L_{p+1}.$$

By Theorem 6.1, $\mathcal{H} \hookrightarrow \mathcal{Q}$ is a monomorphism of simplicial real vector spaces.

First, $\mathcal{H}_p = 0$ for $p \leq 8$. Indeed, (15) gives the claim at label number 9. For every smaller label number, insert zero columns up to nine labels and then apply the corresponding face maps. The face–degeneracy identities give left inverses to the insertion, while Theorem 5.1 guarantees that the faces preserve L. Hence any nonzero class in K_r/L_r for $r < 9$ would produce a nonzero class in K_9/L_9, a contradiction.

Apply the exact normalization functor. Since $\mathcal{H} \hookrightarrow \mathcal{Q}$, exactness gives

$$N(\mathcal{H}) \hookrightarrow N(\mathcal{Q}).$$

For $p \leq 8$ the source $\mathcal{H}_p$ is already zero, so $N(\mathcal{H})_p = 0$. For $p \geq 9$, (14) gives $N(\mathcal{Q})_p = 0$, and hence $N(\mathcal{H})_p = 0$ as well. Thus

$$N(\mathcal{H}) = 0$$

in every degree. Faithfulness of normalization (equivalently, Dold–Kan) forces $\mathcal{H} = 0$. Therefore $K_n/L_n = 0$ for every $n \geq 6$. □

8.1 An all-n quotient-rank formula

The normalized decomposition of a simplicial vector space is indexed by order-preserving (weakly increasing) surjections $[n-1] \twoheadrightarrow [p]$, where $[p] = \{0, \dots, p\}$. There are $\binom{n-1}{p}$ such surjections. Using Proposition 7.1, one obtains

Corollary 8.2. *For every n,*

$$\dim(V_n/L_n) = \binom{n-1}{5} + 9\binom{n-1}{6} + 19\binom{n-1}{7} + 12\binom{n-1}{8}, \tag{16}$$

where binomial coefficients outside their natural range are zero. Consequently

$$\operatorname{rank} L_n = \dim K_n = |\mathcal{U}_n| - \dim(V_n/L_n),$$

with $|\mathcal{U}_n|$ given by (5)*.*

9 The Magic Number corollary

Proof of Theorem 1.1. Let $\mathcal{T}$ and $\mathcal{T}'$ be complete multiplicity-one all-Z tilings. By (3),

$$\chi_{\mathcal{T}} - \chi_{\mathcal{T}'} \in K_n.$$

By Theorem 8.1, this difference lies in L_n. The augmentation functional

$$\epsilon : V_n \to \mathbb{R}, \qquad \epsilon\left(\sum_S c_S e_S\right) = \sum_S c_S$$

vanishes on L_n by (6). Therefore

$$|\mathcal{T}| - |\mathcal{T}'| = 0.$$

All complete all-Z tilings thus have the same number of tiles.

A standard BCFW tiling exists for every admissible n and has the Narayana cardinality

$$\frac{1}{n-3}\binom{n-3}{2}\binom{n-3}{3}$$

for $k = 2$, by the $m = 4$ BCFW tiling theorem and the established BCFW count [4, 5]. Comparing an arbitrary tiling to a standard BCFW tiling gives the stated formula. The algebraic simplification to $(n-3)(n-4)^2(n-5)/12$ is immediate. □

Remark 9.1 (Scope)**.** *The theorem concerns constant-coefficient all-Z indicator relations and complete all-Z positroid tilings. It does not assert that the full fixed-Z indicator kernel equals L_n for every individual positive Z, that the flip graph is connected, or that L_n is spanned by actual global tiling differences for arbitrary n. It also makes no claim for $k > 2$.*

10 Discussion

The proof separates into a rank-two local classification and a global simplicial argument. The local step is strongly rank-specific: it uses the description of $\mathrm{Gr}_{\geq 0}(2, n)$ by cyclic parallel classes and a finite classification of nine-dimensional parents. By contrast, the final mechanism—adjacent amalgamation, loop insertion, normalization, and support-bounded vanishing—is formulated at the level of source-cell relations and may admit higher-rank analogues. Determining which part of this structure survives for $k \geq 3$ is a natural next problem toward the full $m = 4$ Magic Number Conjecture.

The proof also contrasts with the $m = 2$ argument of [7]. There, a Parke–Taylor weight gives a direct unit valuation on tiles. Here the augmentation functional becomes well-defined only after the global relation theorem $K_n = L_n$. The finite $n = 8, 9$ indicator certificates do admit signed rational unit functionals (`N8-UNIT`, `N9-UNIT`), but their specific weights do not appear to depend only on simple tile profile data (the finite probes are `WEIGHT-PROBES`). A canonical $m = 4$ weight formula, if one exists, would provide an alternative route to higher rank.

A Exact finite certificates

The supplement is a directory named `publication/`. Paths in this appendix are relative to that directory. A certificate ID denotes a finite mathematical claim, its literal data, and the checks needed to validate it. These are distinct from software regression tests. The complete claim-to-file mapping is `CERTIFICATE_INDEX.md`; the source of each check is `verifier/checks.py`, with exact linear algebra in `verifier/exact.py`. From the supplement directory, one may run

```
python verify_all.py --certificate MEM-CORES
python verify_all.py
```

The first command replays the indicated family and its required inputs; replacing `MEM-CORES` by any ID below selects that family. The second runs the full release verification. A failed check terminates with a nonzero exit status. Only a successful full replay may print `ALL PUBLICATION-CRITICAL CERTIFICATES PASSED`.

Integers and rational numbers are stored exactly; a rational string `a/b` means $a/b \in \mathbb{Q}$. Matrix rows and columns have the order given in their files. Indices into arrays are zero-based, while source labels and the suffixes of public IDs are one-based. Determinant signs always refer to the published row and column order. Stable IDs, rather than a display order chosen by a reader, identify the certificates.

A.1 Membership catalogue and exclusions

`MEM-CORES`. The literal catalogue is `certificates/membership/catalogue.json`. Each record gives an ID `M001`, etc., the nonloop support q, the number s of ordered cyclic parallel classes, the classes themselves, their sizes, the nonzero bases, the affine-permutation label, dihedral orbit representative and stabilizer, membership status, and the applicable mechanism. Excluded cores are retained. The verifier independently enumerates the loopless rank-two cells with $q + s = 12$, compares the full labeled sets and dihedral data, and applies the criterion in Theorem 3.1. The counts of admitted labeled cells are exactly $1, 14, 36, 30, 5$ for $q = 6, \ldots, 10$. Choosing the nonloop support in $[n]$ then gives (5). This check verifies the finite exhaustion and arithmetic; the proof of the classification mechanisms is in Theorem 3.1 and its cited input.

MEM-FOUR-MASS. `certificates/membership/four_mass.json` contains the exact matrices Z, C_1, C_2, G and the four-mass class pattern. The check verifies every increasing six-minor of Z is positive; every source two-minor is nonnegative with exactly the declared zero pattern; the stacked matrix with row blocks C_1, C_2 has rank greater than two; and

$$\det G \neq 0, \qquad C_1 Z = G(C_2 Z).$$

Thus C_1, C_2 represent distinct source planes in the same open positroid cell and have the same image plane for this positive Z. One such counterexample suffices to exclude all-Z injectivity. The original certificate is retained in `certificates/membership/four_mass_historical.json`.

MEM-SPURION. `certificates/membership/spurion_inverse.json` contains exact source, external, and target matrices for the inverse fixtures, together with expected outcomes and their declared scope. The verifier checks external positivity, the source-image identities, and the exact inverse outcomes. The generic inverse used in the local and $n = 9$ certificates is specified next. Its finite fixtures are not an all-Z injectivity proof; the published spurion assertion cited in Theorem 3.1 is the uniform geometric input.

A.2 Exact membership decisions and open witnesses

The implementation is `independent/membership.py`. Its input is an exact rational positive $n \times 6$ matrix Z, a rank-two target frame Y, and an admissible source label S. The function `prepare` removes loops and chooses a signed cyclic chart. The decision function `decide` returns 1 exactly for $Y \in f_Z(S)$ under the generic conditions below. It returns 0 for a strictly excluded target; an exceptional equality or rank condition causes an error, not a silent nonmembership answer. These are decisions for the open source image. The generic conditions ensure the same indicator value holds on an open target neighborhood, as required for relations modulo dimension seven.

Complement-five inverse. The distinguished class D is placed last. The five complementary rows span H, and the routine `complement` computes the line $Y \cap H$ in the basis of the first six external rows. It requires $Y \not\subset H$ and all five line coordinates to be nonzero. Mixed signs exclude a positive source. Otherwise their common sign is removed and a common scale fixed, giving $a_i > 0$. For each complementary class A_j, form

$$V_j = \sum_{i \in A_j} a_i Z_i.$$

Choose a second target row with positive h-value and expand it in the six-vector basis

$$(V_1, \ldots, V_\ell, Z_{D_1}, \ldots, Z_{D_r})$$

from the proof of Theorem 3.1. The coefficients give slopes t_j and entries d_h. The exact source reconstructed from these coefficients is checked against the target up to its explicitly computed invertible row change. The routine requires the consecutive differences $t_{j+1} - t_j$ and all d_h to be nonzero. It tests the signs of all source two-minors and, when admitted, their precise zero/nonzero pattern. Equivalently in this chart, admission means $t_1 < \cdots < t_\ell$ and every $d_h > 0$. A negative minor of the unique candidate excludes membership. The inverse argument in Theorem 3.1 proves necessity and sufficiency of these checks, rather than merely giving a sufficient numerical test.

Spurion inverse. In the signed chart with three consecutive classes A, B, D of size three, write a candidate as

$$C(x) = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_4 & x_5 & x_6 & x_7 & x_8 & x_9 \end{pmatrix}.$$

The routine `spurion` solves the twelve homogeneous ambient coordinates of $C(x)Z = GY$ in the nine x_i and four entries of G. On the declared generic set, this 12×13 system has a one-dimensional nullspace, each x_i is nonzero, and the stack of Y with each external class triple has rank five. The check admits the target exactly when the nine x_i have a common sign, which may be reversed. It also verifies equality of image planes. Indeed, three distinct cyclic source directions can be put in the displayed gauge, and positivity is precisely the common-sign condition. The unique nullspace therefore supplies both an inverse test and a nonmembership test on this generic set. The ambient formulation shares no production matrix-building routine with the earlier quotient-coordinate inverse.

Openness. The recorded witnesses supply rational target frames and positive source preimages where membership is asserted. The replay checks the relevant nonzero minors, ranks, and strict coefficient signs just described. For the hostile and $n = 9$ witnesses it also checks all increasing four-twistors $\det(Y_1, Y_2, Z_i, Z_j, Z_k, Z_l)$ are nonzero. Nonzero determinants and strict inequalities persist on an open neighborhood. Each matrix entry therefore records a locally constant indicator value; a boundary sample is rejected. This is why these rows constrain K_n, whose definition ignores lower-dimensional sets.

A.3 Parent cores and local relations

`LOCAL-PARENTS` and `LOCAL-RELATIONS`. `certificates/local/parents.json` and its readable companion `certificates/local/PARENTS.md` give the 29 cores `P001`,...,`P029`. Each records its ordered classes, facet IDs, facet membership mechanisms, sector, relation dimension, and full primitive coefficient vector in the published facet order. Positive and negative facet-ID lists give the literal sides of every $3 \leftrightarrow 3$ or $5 \leftrightarrow 5$ relation. The verifier enumerates the types with $q + s = 13$, reconstructs their facets, compares every facet label, checks the partition $12 + 10 + 5 + 2 = 29$, and verifies every coefficient sum is zero. The original signed vectors are retained in `certificates/local/relations_historical.json`. The geometric proof in Theorem 4.1 establishes that these vectors are all-Z relations; finite sign-vector arithmetic alone does not do so.

`LOCAL-TEN-WALL`. `certificates/local/overlap_witnesses.json` gives the 45 witnesses, with parent and witness IDs. The five original family records, including ordered walls and edge matrices, are `certificates/local/overlaps_historical.json`. Each witness gives Z, Y, two exact positive preimages, source coefficients, and image row changes. The verifier checks these identities and uses the exact inverse on all ten facets, including the eight asserted nonmembers. Its row has ones at the two present facets and zeros elsewhere, yielding the coefficient equation $c_i + c_j = 0$. The nine rows form a 9×10 matrix; deleting the specified vertex column gives a minor of absolute determinant one. The known balanced sign vector lies in its kernel. Consequently the kernel is one-dimensional. This is the finite upper-bound certificate for each ten-wall parent relation space.

`LOCAL-HOSTILE-8` and `LOCAL-HOSTILE-9`. The literal files `certificates/local/hostile_q8.json` and `certificates/local/hostile_q9.json` contain the ordered eight admissible facets, external matrix, eight target records, and indicator matrix for each hostile parent. Replaying all

64 decisions in each file reconstructs its 8×8 matrix; the exact determinant in the published order must equal 1. This excludes every nonzero coefficient relation on those eight facets.

A.4 The complete $n = 9$ certificate

Universe and local rank. `N9-UNIVERSE` is `certificates/n9/tiles.json`: the complete ordered list `N9-T0001`,...,`N9-T0942`. The independent enumeration must equal this list exactly, not merely have the same size. `N9-LOCAL-RANK` is `certificates/n9/R9.json`, whose 693×942 integer matrix R_9 is reconstructed from the transported local prototypes. The file includes row and column IDs, selected independent rows, pivot columns, and sparse rational reduced rows. The selected 470×470 minor has determinant -1, proving rank at least 470. For the upper bound, write B for the 470 published reduced rows and $p_1, \ldots, p_{470}$ for their pivot positions. The verifier checks $B_{i,p_j} = \delta_{ij}$ and, for every original row r, the exact identity

$$r = \sum_{j=1}^{470} r_{p_j} B_j.$$

Thus every row lies in a 470-dimensional space and the rank is exactly 470. A nonzero minor alone would not prove this upper bound.

Witnesses and indicator rank. `N9-INDICATORS` uses `certificates/n9/witnesses.json`, with a common exact external matrix and target IDs `N9-W0001`,...,`N9-W0472`. The source frames, target frames, source labels, and recorded indicator rows suffice to reconstruct every entry by Appendix A.2. The independently rebuilt 472×942 binary matrix must agree entrywise with both `certificates/n9/A9.json` and the original `certificates/n9/indicator_historical.json`. All $472 \cdot 942 = 444{,}624$ decisions are replayed; none is inferred from agreement on a sample.

For `N9-INDICATOR-RANK`, `certificates/n9/A9.json` identifies the 472 selected column indices in their determinant order; its row IDs identify all 472 selected rows. The verifier extracts that literal submatrix from the reconstructed matrix and checks its determinant is 4. The lower bound is therefore 472, and the number of rows is the matching upper bound.

Annihilation and conclusion. `N9-ANNIHILATION` checks every entry of $A_9 R_9^{\mathsf{T}}$ is zero by exact integer arithmetic. `N9-KERNEL` combines the preceding checks with the geometric inclusion $L_9 \subseteq K_9$ and the open-witness implication $K_9 \subseteq \ker A_9$:

$$\dim L_9 = 470, \qquad \dim \ker A_9 = 942 - 472 = 470, \qquad L_9 \subseteq K_9 \subseteq \ker A_9.$$

It follows that $K_9 = L_9$. The finite matrix kernel is used as an upper bound on the all-Z relation space; no extrapolation from one Z to all positive Z is made.

A.5 Normalized quotients and support cutoff

The six literal files `certificates/normalized/N6.json`,..., `certificates/normalized/N11.json` are `NORM-6`,..., `NORM-11`. Each contains canonical column IDs and source labels, row IDs and parent labels, the projected integer matrix, selected rows and pivot columns, sparse rational reduced rows, and the exact selected minor determinant. The construction retains exactly the tile columns with no loop in positions $2, \ldots, n$. Each transported parent relation is projected onto those columns, with repeated facet images combined. The reconstructed rows and labels must agree with the file in full.

The six dimensions, row counts, ranks, and quotient dimensions must equal Proposition 7.1. The verifier uses the same exact row-span upper certificate and selected-minor lower certificate as for R_9. In particular, upper certificates are required at $n = 7, 8, 9$. At $n = 10$ the 35×35 selected minor has determinant -1; at $n = 11$ the 5×5 selected minor has determinant 1. At $n = 6$ there are no rows and the rank is zero; the empty determinant is 1 by convention and asserts no positive rank.

`SUPPORT-CUTOFF` ties the catalogue support $q \leq 10$ to the normalization convention and checks that the normalized column set is empty at $n = 12$. The statement for all $n \geq 12$ is the symbolic pigeonhole argument of Proposition 7.1: with at most ten nonloops there must be a loop in positions $2, \ldots, n$. The label number n always corresponds to simplicial degree $p = n - 1$.

A.6 Additional finite observations

The Discussion's finite unit-function observations have separate IDs `N8-UNIT`, `N9-UNIT`, and `WEIGHT-PROBES` in `CERTIFICATE_INDEX.md`. The first two expose the original rational weights λ and indicator matrices A and check $A^{\mathsf{T}}\lambda = \mathbf{1}$ exactly. The third exposes the recorded finite comparisons of weights with tile-profile data. These are supplementary finite observations, not premises of the all-n theorem or a claim of nonexistence of a general weight formula.

B Reproducibility and dependency map

Table 2 separates the finite checks from the uniform geometric arguments. The supplement's `PROOF_DEPENDENCIES.md` records the finer claim-to-certificate links. An edge labelled mixed uses both a symbolic argument and a finite certificate; numerical replay does not verify the symbolic argument or an external theorem.

Input	Consequence	Dependency type
Membership mechanisms and `MEM-CORES`	Support bound and parent-facet classification	Mixed; spurion input external
`LOCAL-PARENTS`, `LOCAL-RELATIONS`, overlap and hostile certificates	Local theorem and $L_n \subseteq K_n$	Mixed
Amalgamation, zero insertion, and truncation	Simplicial $V, L, K, K/L$	Human/symbolic
Local theorem and the `N9-*` certificates	$K_9 = L_9$	Mixed
`NORM-10`, `NORM-11`, support bound	$N(\mathcal{Q})_p = 0$ for $p \geq 9$	Mixed
Simplicial structure, base case, normalized vanishing, Dold–Kan	$K_n = L_n$ for every $n \geq 6$	Human/symbolic and external citation
Balanced local vectors, global generation, standard BCFW tiling	Magic Number cardinality	Mixed and external citation

Table 2: Proof dependencies. Monospaced certificate IDs mark the machine-assisted inputs; the finite lower normalized ranks additionally give Corollary 8.2.

What independent reconstruction means here. The historical generators and their outputs remain in the research repository. The release copies retain provenance rather than replacing those artifacts. The standalone reconstructor in `independent/` does not import the production enumeration, facet, transport, or membership functions from `src/`. It reconstructs the finite labeled sets, projected matrices, and witness decisions from the published conventions and certificate inputs. Its membership component adapts the historical secondary ambient inverse verifier; it is not a third independently invented inverse. The release catalogue normalizer also shares the standalone combinatorial routines, so independence refers to comparison with the historical producers. The implementations share the mathematical classification and signed local prototypes, and use common exact arithmetic libraries, including SymPy. Their agreement is an independent implementation check, not a separately proved classification theorem or an independently implemented computer algebra system. The supplement distinguishes this reconstruction from replay of existing literal data and from software regression tests.

Reproducibility files. `REPRODUCIBILITY.md` records the tested interpreter and platform, pinned dependencies, installation and clean replay commands, resource requirements, and cache behavior. Verification uses exact arithmetic and requires no network after dependency installation. The manifest `SHA256SUMS.txt` fixes the manuscript, PDF, literal certificates, verifier sources, and release records. `SUBMISSION_READINESS.md` reports the actual release-validation state; the candidate designation here does not assert that an archive has been deposited. The historical symbolic notes are included under `proofs/`, notably `proofs/research/run4/LOCAL_PARENT_THEOREM.md`, `proofs/research/run6/MEMBERSHIP_CORES.md`, `proofs/research/run6/SPURION_INVERSION.md`, `proofs/research/run6/N9_INDICATOR_KERNEL.md`, and the run7 amalgamation, simplicial, and global-generation notes.

Code and data availability. The accompanying supplement is the candidate release `v1.0.0-paper`. Its historical base commit is `284c19a88a8e1327f139d4e3c58d16c751610b73`; a release commit has not yet been minted. The intended public repository is [**PUBLIC REPOSITORY URL TO BE INSERTED**], and the archival identifier is [**ZENODO DOI TO BE INSERTED AFTER RELEASE**]. Neither placeholder denotes an existing upload. From the deposited supplement's root, the verification command is `python verify_all.py`. Code is offered under the MIT license; the manuscript and data are offered under CC BY 4.0.

AI assistance disclosure

OpenAI ChatGPT and Codex models assisted with proof exploration, finite-case formalization, code generation, verification tooling, adversarial proof review, and manuscript drafting and editing. AI systems are not authors. The human author remains responsible for all mathematical claims, computations, citations, and the final manuscript.

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